WORKED SOLUTION

STEP III 2023, Question 1

Pure • Tangents to Circles and Parabolas

STEP III 2023, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

A line is tangent to a circle exactly when its perpendicular distance from the centre equals the radius. So we write down the chord PQ in terms of p and q, compute that distance, and watch the tangency condition collapse to the given relation. After that, everything is about a single symmetric equation in two parameters: the final part asks whether the relation still holds for the *third* side, which is a job for sums and products of roots.

Solution

Part (i)

The gradient of PQ is

$$\frac{aq^2 - ap^2}{2aq - 2ap} = \frac{p+q}{2},$$

dividing by $q - p \neq 0$ because the points are distinct. The line through P is therefore $y - ap^2 = \frac{1}{2}(p+q)(x - 2ap)$, which simplifies to

$$(p+q)x - 2y - 2apq = 0.$$

The distance from the centre $C(0, 3a)$ to this line is

$$\frac{|-6a - 2apq|}{\sqrt{(p+q)^2 + 4}} = \frac{2a|pq+3|}{\sqrt{(p+q)^2+4}}.$$

The line is tangent to the circle exactly when this equals $2a$, that is (squaring, since both sides are positive, and cancelling $4a^2 \neq 0$) when

$$(pq+3)^2 = (p+q)^2 + 4 \iff p^2q^2 + 6pq + 5 = (p+q)^2.$$

This is $(*)$, so when $(*)$ holds the line PQ is a tangent to the circle. (Every step was reversible, so the converse holds too, and we shall need that in part (iii).)

Part (ii)

Expanding $(*)$ and collecting powers of q:

$$(1-p^2)q^2 - 4pq + (p^2-5) = 0.$$

Since $p^2 \neq 1$, the leading coefficient is nonzero, so this is a genuine quadratic in q. Its discriminant is

$$\begin{aligned} 16p^2 - 4(1-p^2)(p^2-5) &= 16p^2 - 4(-p^4+6p^2-5) \\ &= 4(p^4 - 2p^2 + 5) \\ &= 4((p^2-1)^2 + 4) > 0, \end{aligned}$$

so there are two distinct real roots q_1, q_2. Neither equals p: putting $q = p$ in the left-hand side gives $-p^4 - 2p^2 - 5 < 0$. So Q_1 and Q_2 really are points distinct from P.

From the quadratic,

$$q_1 + q_2 = \frac{4p}{1-p^2}, \qquad q_1 q_2 = \frac{p^2-5}{1-p^2}.$$

Part (iii)

Take the triangle PQ_1Q_2, where $Q_i = (2aq_i, aq_i^2)$ and q_1, q_2 are the roots from part (ii). Its vertices all lie on $x^2 = 4ay$, and they are three distinct points (as shown in (ii)); three distinct points of a parabola are never collinear, since a line meets a parabola at most twice. So this is a genuine triangle.

By part (i), the sides PQ_1 and PQ_2 are tangent to the circle, since (p, q_1) and (p, q_2) satisfy $(*)$.

Ask yourself How can we show that the third side Q_1Q_2 is tangent when we don't want to solve for q_1 and q_2 explicitly?

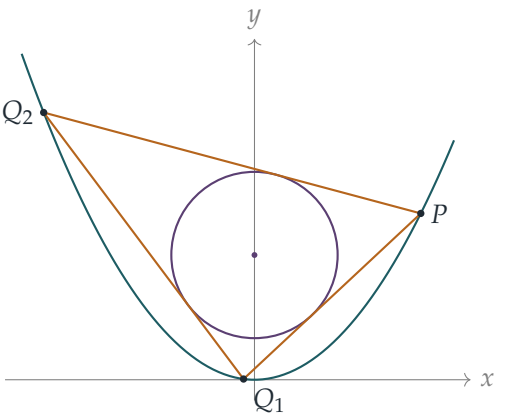
By the reversible argument of part (i) (with p, q replaced by q_1, q_2), Q_1Q_2 is tangent precisely when

$$(q_1+q_2)^2 = q_1^2 q_2^2 + 6q_1q_2 + 5,$$

and both sides involve only $s = q_1 + q_2$ and $t = q_1 q_2$, which we know. Multiply through by $(1-p^2)^2$. The left-hand side becomes $16p^2$. The right-hand side becomes

$$\begin{aligned} &(p^2-5)^2 + 6(p^2-5)(1-p^2) + 5(1-p^2)^2 \\ &\quad = (p^4 - 10p^2 + 25) + (-6p^4 + 36p^2 - 30) + (5p^4 - 10p^2 + 5) \\ &\quad = 16p^2. \end{aligned}$$

The two sides agree, so Q_1Q_2 is also tangent to the circle, and PQ_1Q_2 is the required triangle.



The figure shows the case $a = 1$, $p = 2$, where $q_{1,2} = \frac{1}{3}(-4 \pm \sqrt{13})$.

Checks

- Take $a = 1$, $p = 2$: the quadratic is $3q^2 + 8q + 1 = 0$, with $q_1 + q_2 = -\frac{8}{3} = \frac{4p}{1-p^2}$ and $q_1 q_2 = \frac{1}{3} = \frac{p^2-5}{1-p^2}$. Then $(q_1 + q_2)^2 = \frac{64}{9}$ and $q_1^2 q_2^2 + 6q_1 q_2 + 5 = \frac{1}{9} + 2 + 5 = \frac{64}{9}$, as part (iii) requires.
- Numerically, with $q_1 \approx -0.1315$, $q_2 \approx -2.5352$, all three side lines are at distance 2.000 from $(0, 3)$; a computer check over thousands of random a and p agreed to about 10^{-13}.
- The excluded case $p = \pm 1$ is instructive: for $p = 1$ the quadratic degenerates to $-4q - 4 = 0$, giving only $q = -1$. The "missing" tangent from $P(2a, a)$ is the vertical line $x = 2a$, which touches the circle but meets the parabola only at P.

Common slips

- Forgetting the modulus in the distance formula, or squaring without noting that both sides are positive.
- Using only the "if" direction of part (i): part (iii) needs the converse to turn tangency of Q_1Q_2 into an algebraic condition, so say that the steps are reversible.
- Not checking that $q_1, q_2 \neq p$, so that the triangle is not degenerate.

Beyond STEP

This is an instance of *Poncelet's porism*: if a triangle can be inscribed in one conic and circumscribed about another, then such a triangle can be started from *any* point of the first conic. Here the parabola and the circle are such a pair, and part (iii) proves closure

directly with Vieta's formulas. The general proof uses projective geometry and, most elegantly, elliptic curves. Siklos's *Advanced Problems in Mathematics* has more questions in this style of parametric conic geometry.

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